

Unit 4, Lesson 2: Identifying Key Features of Lines From Graphs



Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



Learning Target

- ☐ I can identify key features of a linear function from a graph.



4.2.1 Warm-Up: What Do You Notice?

1. What do you notice about the equation $y + 7 = \frac{5}{4}(x - 8)$ and the function $f(x) = \frac{5}{4}x - 1$?
2. If you substitute the same x -value, or input, into the equation and function, do you think that the equation and function will give the same outputs given the same input?
3. What do you notice about the equation $y - 3 = -6(x + 5)$ and the function $f(x) = -6x - 27$?
4. Do you think the equation and function will give the same outputs given the same input?

5. Substitute the x -value in the equation and the function to determine if the output is the same.

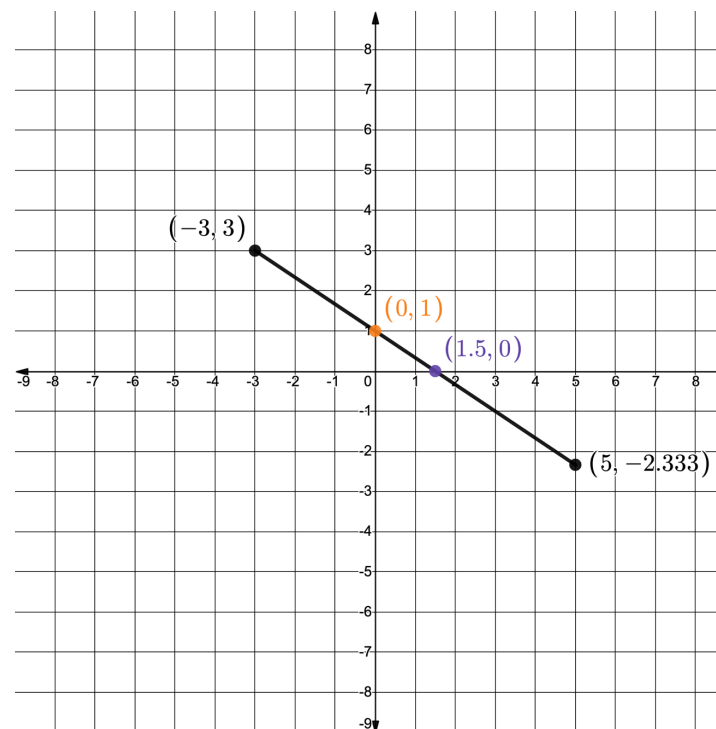
x -value	Equation	Function
-2	$y + 7 = \frac{5}{4}(x - 8)$	$f(x) = \frac{5}{4}x - 1$
9	$y - 3 = -6(x + 5)$	$f(x) = -6x - 27$

6. What is one observation you have about the equation and function that do **not** have the same output?
7. What is one observation you have about the equation and function that do have the same output?



4.2.2 Exploration: Key Features From a Graph

1. A line segment is shown on the coordinate plane. The line segment represents a function.



- a. Draw a vertical line from the left endpoint of the line segment to the x -axis. What is the x -value?
- b. Draw a vertical line from the right endpoint of the line segment to the x -axis. What is the x -value?

- c. Discuss with your partner how finding the x -values of the left endpoint and the right endpoint of the line segment helps to determine the domain of the function the line segment represents.

- d. Use the x -values of the left endpoint and the right endpoint of the line segment to write the domain as an inequality.

$$\underline{\hspace{2cm}} \leq x \leq \underline{\hspace{2cm}}$$

- e. Work with your partner to determine the range of the function that is represented by the line segment.

- f. Use the y -values of the left endpoint and the right endpoint of the line segment to write the range as an inequality.

$$\underline{\hspace{2cm}} \leq y \leq \underline{\hspace{2cm}}$$

- g. Work with your partner to find the other key features of the line segment.

- The slope of the line segment is _____.
- As the x -values are increasing, the y -values are _____.
- The coordinates of the point where the line crosses the y -axis are _____. The y -value of the point is _____.
- The coordinates of the point where the line crosses the x -axis are _____. The x -value of the point is _____.

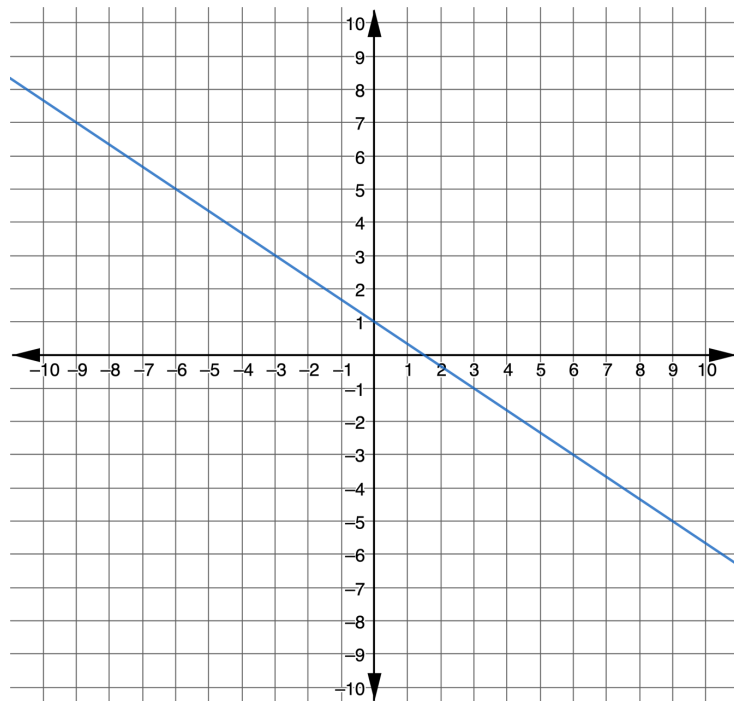


The **intercept** is the value of a variable when all other variables in the equation equal 0. On a graph, the intercept is the value where a function crosses an axis.

- h. Work with your partner to use the definition of **intercept** to determine the intercepts.

x -intercept	y -intercept

2. A line that represents a function is shown on the coordinate plane. The line is similar to the line segment represented in the previous problem except it does not have endpoints.



Work with your partner to find the key features of the line.

Domain	Range	As x -values increase, y -values ...
Slope	x -intercept	y -intercept

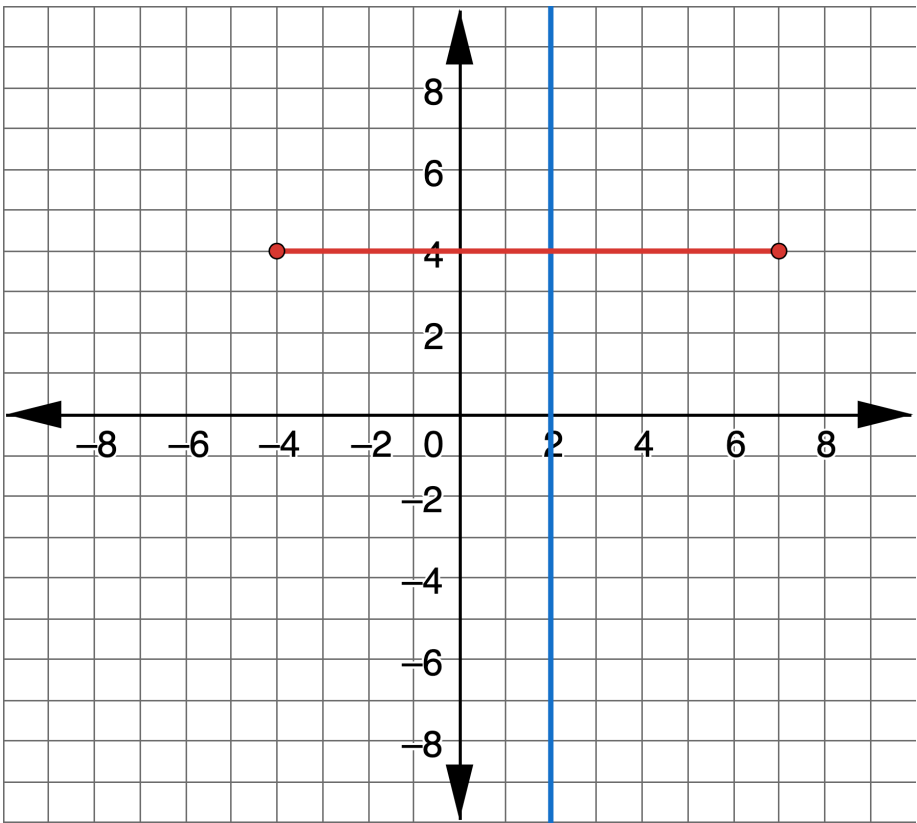
3. Which key features were different for the line and line segment?





4.2.3 Exploration: Horizontal and Vertical Lines

1. The graph shows a vertical line in blue and a horizontal line segment in red.



Work with your partner to find the key features for each.

Red Line Segment		
Domain	Range	As x -values increase, y -values ...
Slope	x -intercept	y -intercept

Blue Line		
Domain	Range	As x -values increase, y -values ...
Slope	x -intercept	y -intercept

2. How does the range of a horizontal line relate to its y -intercept?
3. How does the domain of a vertical line relate to its x -intercept?



4.2.4 Wrap-Up: Reflection

Complete each statement as you reflect on today's learning.

1. The concept that made the most sense to me today is...

2. One concept that I still do not understand from today is...

Unit 4, Lesson 3: Identifying Key Features of Lines From Tables and Equations



Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



Learning Targets

- ☐ I can identify key features of a linear function from a graph.
- ☐ I can identify key features of a linear function given a table or an equation.